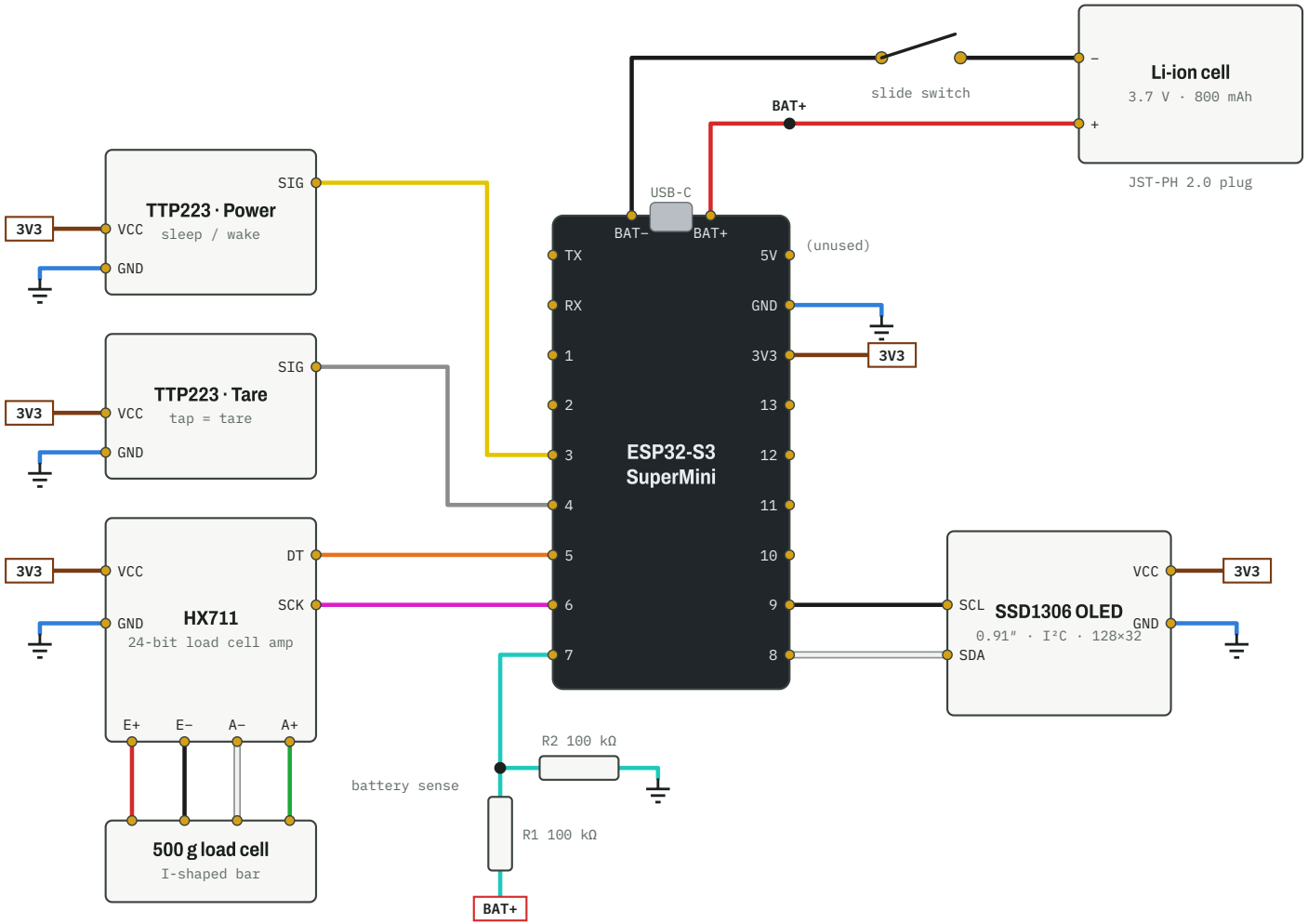
ESP32-S3 SUPERMINI BUILD · FIRMWARE V2.2.1

WeighMyBru² SuperMini Wiring

Every connection between the ESP32-S3 SuperMini, HX711, load cell, OLED, two TTP223 touch modules, and the Li-ion battery. Pin numbers come from `include/BoardConfig.h`. Wire colors match the official diagram on weighmybru.com. Hover a row in the table to highlight its wire.



● Dot = joined wires. Crossing lines without a dot are not connected. **3V3** flag = ESP32 3V3 pin ⏚ = ESP32 GND

Pin map

ESP32 PIN	GOES TO	WIRE	FIRMWARE NAME
GPIO3	TTP223 Power · SIG	yellow	TOUCH_SLEEP_PIN
GPIO4	TTP223 Tare · SIG	gray	TOUCH_TARE_PIN
GPIO5	HX711 · DT	orange	HX711_DATA_PIN
GPIO6	HX711 · SCK	magenta	HX711_CLOCK_PIN
GPIO7	Divider midpoint (R1/R2)	cyan	BATTERY_PIN
GPIO8	OLED · SDA	white	I2C_SDA_PIN
GPIO9	OLED · SCL	black	I2C_SCL_PIN
3V3	VCC on HX711, OLED, both TTP223	brown	
GND	GND on HX711, OLED, both TTP223, R2	blue	
BAT+	Battery + and R1	red	
BAT−	Slide switch, then battery −	black	

Load cell to HX711

LOAD CELL WIRE	HX711 PAD
Red	**E+**
Black	**E−**
White	**A−**
Green	**A+**

Build notes

The switch cuts the negative lead

Battery − runs through the slide switch to the board's BAT− pad. Battery + goes straight to BAT+. Use the JST-PH 2.0 connector on the battery leads so you can unplug the battery.

Battery sense divider

Two 100 kΩ resistors split BAT+ in half for GPIO7. The firmware multiplies the reading by 2.0 (`VOLTAGE_DIVIDER_RATIO`), so a full 4.2 V cell reads 2.1 V at the pin. Use 1% resistors, or the battery percentage drifts.

Touch pads are TTP223 modules

The firmware reads GPIO3 and GPIO4 with `digitalRead` and an internal pull-down. Each pad needs a TTP223 module with SIG, VCC and GND. A bare copper pad will not work.

Both touch pins do extra jobs

GPIO3 wakes the scale from deep sleep. Holding the tare pad (GPIO4) during boot clears the saved WiFi credentials.

Everything runs on 3.3 V

HX711, OLED and both touch modules take power from the 3V3 pin. Leave the 5V pin unconnected.

Negative weight readings?

Swap the white and green load cell wires on A− and A+.

XIAO ESP32S3

The firmware uses the same GPIO numbers on the XIAO, but the pins sit in different physical spots. See `wiring-diagram-xiao.pdf` for that board.

Sources: `include/BoardConfig.h`, `include/BatteryMonitor.h`, `src/main.cpp`, and the official assembly guide.